C# 1D Array Operations

Complete code collection • Beginner-friendly examples

Note: Each example is a separate complete C# program. For insertion examples, the array has one extra slot. Deletion examples use `size` to track active elements.

1. Array Traversal

```
using System;

class Program {     static void Main()     {
int[] numbers = { 11, 22, 33, 44, 55 };

        for (int i = 0; i < numbers.Length; i++)
{
            Console.WriteLine(numbers[i]);
        }
    }
}
```

2. Array Insertion at the End

```
using System;

class Program {     static void
Main()     {         int[] numbers
= new int[5];

        numbers[0] = 11;
numbers[1] = 22;
numbers[2] = 33;
numbers[3] = 44;

        int newValue = 55;

        numbers[4] = newValue;

        for (int i = 0; i < 5; i++)
{
            Console.WriteLine(numbers[i]);
        }
    }
}
```

3. Array Insertion at the Beginning

```
using System;

class Program {     static void
Main()     {         int[] numbers
= new int[5];

        numbers[0] = 11;
numbers[1] = 22;
numbers[2] = 33;
numbers[3] = 44;

        int newValue = 5;

        for (int i = 3; i >= 0; i--)
{             numbers[i + 1] =
numbers[i];         }

        numbers[0] = newValue;

        for (int i = 0; i < 5; i++)
{
            Console.WriteLine(numbers[i]);
        }
    }
}
```

4. Array Insertion at a Specific Position

```
using System;

class Program

 {     static void Main()     {
int[] numbers = new int[5];

        numbers[0] = 11;
numbers[1] = 22;
numbers[2] = 33;
numbers[3] = 44;

        int position = 3;
int newValue = 25;

        for (int i = 3; i >= position - 1; i--)
{             numbers[i + 1] = numbers[i];
}

        numbers[position - 1] = newValue;

        for (int i = 0; i < 5; i++)
{
            Console.WriteLine(numbers[i]);
        }
    }
}
```

5. Array Deletion at the End

```
using System;

class Program

{     static void Main()     {
int[] numbers = new int[5];

        numbers[0] = 11;
numbers[1] = 22;
numbers[2] = 33;
numbers[3] = 44;

        int size = 4;

        // Display the array
for (int i = 0; i < size; i++)
{
            Console.WriteLine(numbers[i]);
        }
    }
}
```

6. Array Deletion at the Beginning

```
using System;

class Program

{     static void Main()     {
int[] numbers = new int[5];

        numbers[0] = 11;
numbers[1] = 22;
numbers[2] = 33;
numbers[3] = 44;

        int size = 4;

        // Delete at the beginning
for (int i = 0; i < size - 1; i++)
{             numbers[i] = numbers[i + 1];
}

        size--;

        // Display the array
for (int i = 0; i < size; i++)
{
            Console.WriteLine(numbers[i]);
        }
    }
}
```

7. Array Deletion at a Specific Position

```
using System;

class Program

{     static void Main()     {
int[] numbers = new int[5];

        numbers[0] = 11;
numbers[1] = 22;
numbers[2] = 33;
numbers[3] = 44;

        int size = 4;
int position = 2;

        // Delete at a specific position
for(int i = position - 1; i < size - 1; i++)
{             numbers[i] = numbers[i + 1];         }

        size--;

        // Display the array
for (int i = 0; i < size; i++)
{
            Console.WriteLine(numbers[i]);
        }
    }
}
```

8. Array Updation

```
using System;

class Program

{     static void Main()     {
int[] numbers = new int[4];

        numbers[0] = 11;
numbers[1] = 22;
numbers[2] = 33;
numbers[3] = 44;

        int position = 2;
int newValue = 35;

        // Update the array
numbers[position - 1] = newValue;

        // Display the array
for (int i = 0; i < numbers.Length; i++)
{
            Console.WriteLine(numbers[i]);
        }
    }
}
```

9. Array Searching (Linear Search)

```
using System;

class Program

 {     static void Main()     {         int[]
numbers = { 11, 22, 33, 44, 55 };

        int searchValue = 33;
bool found = false;

        for (int i = 0; i < numbers.Length; i++)         {
if (numbers[i] == searchValue)             {
Console.WriteLine("Value found at index " + i);
found = true;                 break;             }         }

        if (found == false)
{
            Console.WriteLine("Value not found");
        }
    }
}
```

10. Array Reset

```
using System;

class Program

 {     static void Main()     {
int[] numbers = { 11, 22, 33, 44 };

        // Reset the array

  for (int i = 0; i < numbers.Length; i++)
{             numbers[i] = 0;         }

        // Display the array
 for (int i = 0; i < numbers.Length; i++)
{
            Console.WriteLine(numbers[i]);
        }
    }
}
```